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FNIRSI 菲尼瑞斯

DPS-150

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PORTABLE CNC DC POWER SUPPLY



PDF Manual, simplified. Includes english-only pages, for DPS-150, purchased 2025 Summer by David Malawey

Review Notes 2026.02:

► it's my first portable power supply. essentially an improvement out-of-the-box for output of a controllable DC power with a readout of amperage, wattage.

► Pro: compact compared with benchtop PSU. Quick response from buttons, effectively can perform limiting of current or voltage. Can take an 18v tool battery, reduce to a lower voltage, and hookup to DC actuator.

► Pro: can operate from PD input or DC input, and swap quickly.

► Cons: user needs a dedicated high-powered usbc adapter with PD. my psu shipped with a PD adapter as a kit but they are truly 2 separate products.

Recommend to buy separately & find most competitive PD adapter.

► Cons: cannot reach the claimed 30v max output except with 30v power supply. Most PD adapters max at 20v. **FNIRSI adapter max is 20v**. It is desired to have a DC supply with boost converter inside for higher voltages.

► On Battery: DPS150 can provide up to the peak battery voltage of source. Fully charged Ridgid 18v battery: starts at 21v, allows 21v setting and DPS reduces setting in real-time with battery voltage drop. No interruption of shutoff of output (good performance).

► FNIRSI brand also offers oscilloscopes & other digital instruments, a sign of quality & specialization.

► DPS-150 offers fairly good size of PDF manual. Company expected to prioritize documentation & support to fair degree.

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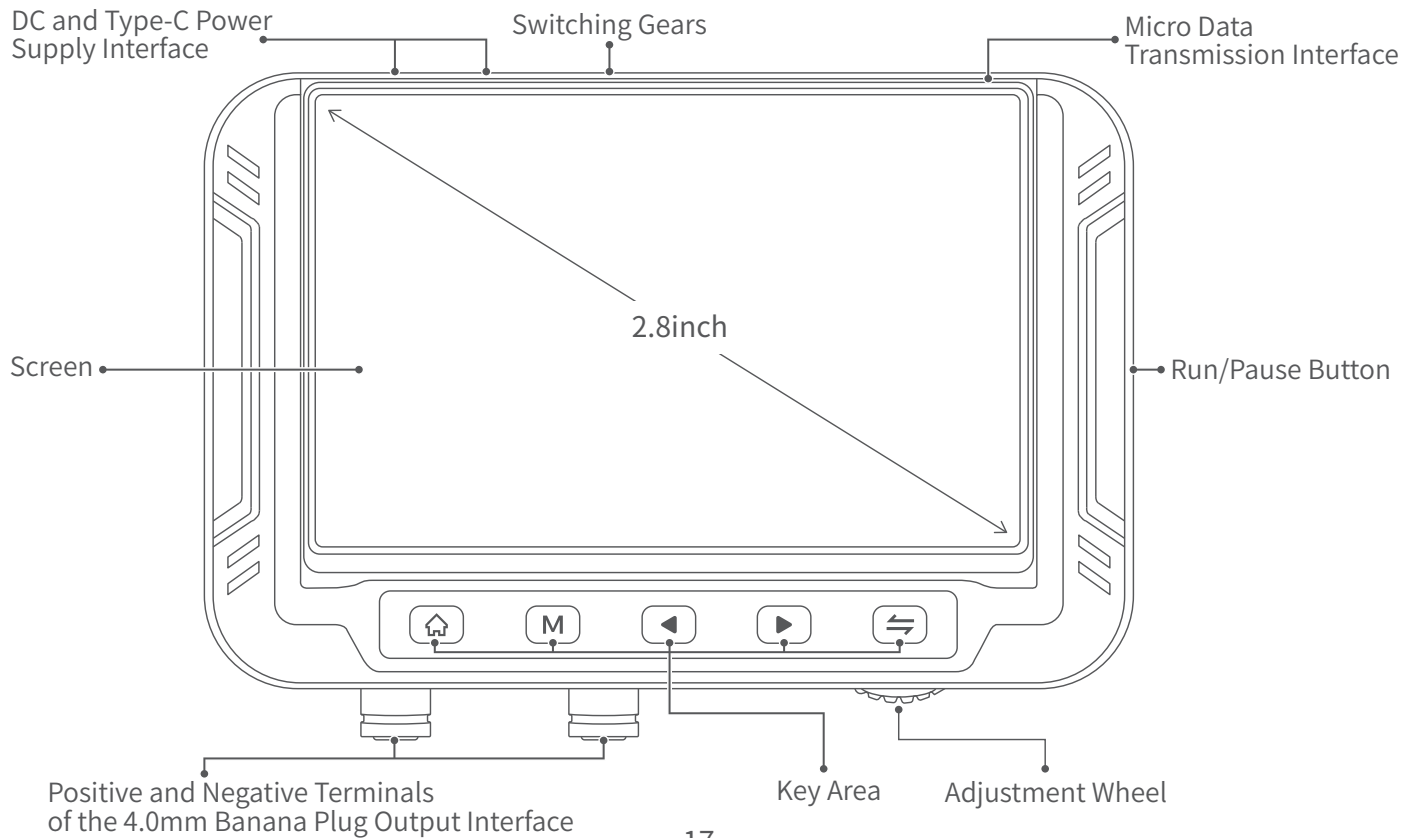
USER NOTICE

- This manual provides detailed instructions on how to use the product and important precautions. Please read this manual carefully and use the product according to the instructions to ensure optimal performance.
- Do not use the instrument in flammable or explosive environments.
- Dispose of used batteries and discarded instruments separately according to national or local regulations; do not dispose of them with household waste.
- If you encounter any quality issues with the instrument or have any questions about its use, please contact "FNIRSI" online customer service or the manufacturer, and we will promptly assist you.

1.PRODUCT INTRODUCTION

The FNIRSI-DSP150 is a high-performance adjustable DC power supply launched by our company. It features a Type-C input interface and multiple power supply modes, allowing precise adjustment of output voltage (0-30V) and current (0-5A). It provides efficient, low-consumption, and stable output, equipped with multiple safety protection functions including overvoltage, overcurrent, overload, overheating, and reverse connection. It can be flexibly applied to serial connection of multiple devices, with rich and user-friendly display and operation, compact and portable humanized design, meeting various application needs. It is designed to provide you with high-performance and stable CNC power supply.

2. PANEL INTRODUCTION



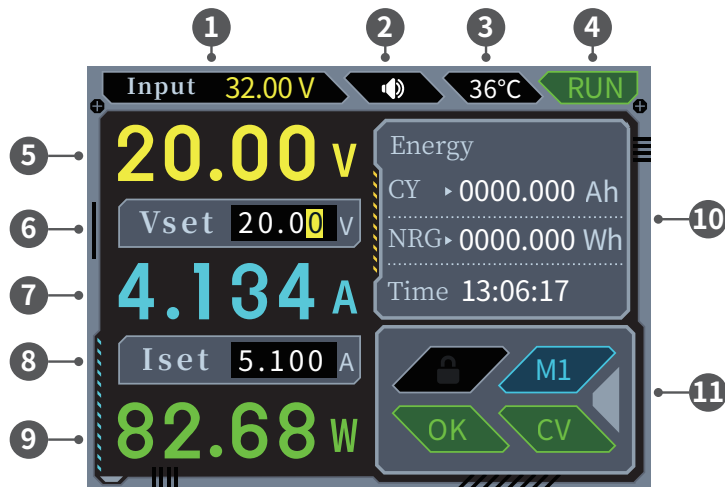
3. PARAMETER INTRODUCTION

Model	DPS-150		Screen	2.8Inch (320*240)	
Input	Voltage	DC5.0V~32V	Output	Voltage	0~30V
	Current	100mA~5A		Current	0~5A
	Supports PD and QC fast charging protocols, power bank			Power	0~150W
Set Resolution	Voltage	10mV	Set Value Precision	Voltage	≤0.1%±5mV
	Current	1mA		Current	≤0.1%±3mA
Readback Value Precision	Input Voltage	≤0.2%±5mV	Load Regulation	0.49%	
	Output Voltage	≤0.1%±10mV	Full Load Efficiency	96.30%	
	Output Current	≤0.1%±5mA (0~3.5A)	Dimensions	106×76×28mm	
Operating Environment	0°C~40°C, 0%~75%RH		Weight	≈178g	
Protection Mechanisms	●Overvoltage Protection ●Overcurrent Protection ●Undervoltage Protection ●Overpower Protection ●Overtemperature Protection ●Reverse Input Protection ●Reverse Output Protection				

4. OPERATION INSTRUCTIONS

Button	Button Type	Interface	Function
	Short Press	/	Run/Pause
	Long Press	/	Lock Button
	Short Press	/	Enter Settings Page
	Long Press	Measurement Switch Off State	Zero Time
		Measurement Switch On State	Zero Capacity, Energy, and Time
M	Short Press	Main Interface	Enter Data Group Page
		Waveform Display Interface	Change Data Group
	Long Press	Waveform Display Interface	Continuously Change Data Group

Button	Button Type	Interface	Function
	Short Press	Main Interface	Switch Data Group
		Waveform Display Interface	Decrease Time Base / Increase Time Base
		Highlight Appears on Setting Page	Left / Right
	Long Press	Highlight Appears on Page	Continuously Cycle Left / Continuously Cycle Right
	Short Press	Main Interface	Highlight Appears on Page
		Settings	Enter Specific Setting Option
	Long Press	/	Switch Between Main Page and Waveform Display Page
	Slip to Left	/	Decrease Value While Adjusting Parameters
	Slip to Right	/	Increase Value While Adjusting Parameters



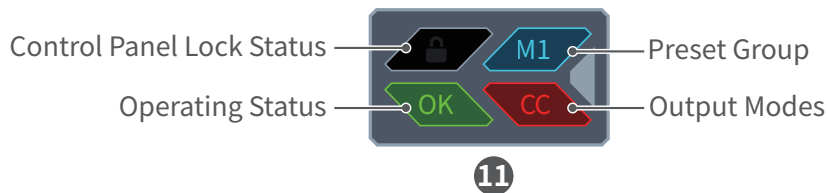
Main Interface

- ① **Input Voltage:** Displays the current input voltage, unit V.
- ② **Sound Icon:** Indicates whether the sound of the device is enabled.
- ③ **Temperature Display:** Shows the internal temperature of the device.
- ④ **Operating Status:** Indicates whether the device is currently running.
- ⑤ **Current Voltage:** Real-time display of the current voltage of the device, unit V.
- ⑥ **Set Output Voltage:** 00.00~30.00V, resolution 0.01V, unit V.

- ⑦ **Current Current:** Real-time display of the current of the device, unit A.
- ⑧ **Set Output Current:** 00.00~5.10A, resolution 0.001A, unit A.
- ⑨ **Current Power:** Real-time display of the current power of the device, unit W.
- ⑩ **Energy Statistics:** Displays the energy (Ah) and capacity (Wh) output by the device.
- ⑪ **Operating Status:** Includes operating status, preset group information, control panel lock status, output mode.

● Operating Status:

- ① Normal OK
- ② Overvoltage Protection OVP
- ③ Overcurrent Protection OCP
- ④ Overpower Protection OPP
- ⑤ Overheat Protection OTP
- ⑥ Reverse Connection Protection REP
- ⑦ Undervoltage Protection LVP



※When any status other than normal is detected, the device automatically shuts off the output and emits a beep alarm. Output is prohibited under undervoltage protection status.

- Preset Group Information: Information about the currently used preset output group, the device supports 6 groups (1-6) of preset groups, where each preset group includes output voltage setting and constant current setting.
- Control Panel Lock Status: Gray indicates unlocked, which turns into a white lock status after being locked. When locked, control panel operations are invalid. It automatically locks when connected to the PC software and cannot be unlocked by pressing buttons.
- Output Modes: There are two types, Constant Voltage Output (CV) and Constant Current Output (CC).

Operation Steps:

Parameter Adjustment:

- ① Press , once to enter parameter adjustment mode. The highlight will appear at the voltage setting. Press  again to cycle between output voltage and current settings.
- ② Use  and the adjustment wheel to adjust the highlighted numerical value by scrolling left or right.

Preset Data Group Operation:

- ① Short press **M** on the main page to enter the preset data group interface.
- ② Press , once to enter parameter adjustment mode. The voltage setting will be highlighted. Press  again to cycle between output voltage and current settings.
- ③ Adjust the highlighted numerical value by scrolling the adjustment wheel left or right using .
- ④ Short press **M** again to save.
- ⑤ Short press  /  on the main page to switch to the previously preset data group.

Operating Status Operations And Control Panel Lock Status Operations:

- ① Short press the  to turn on/off output.
- ② Long press the  to unlock/Lock the keypad panel.

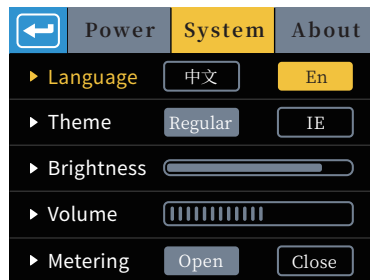
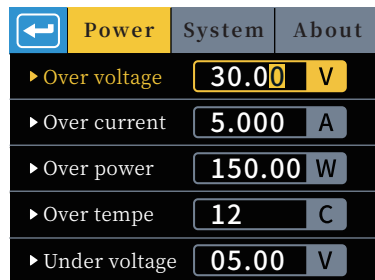
Abnormal Working State Prompt:

- ① Red color indication for abnormal status, including overcurrent protection, overvoltage protection, overpower protection, overheat protection, undervoltage protection, and reverse connection protection, followed by output shutdown.

Menu Interface and Operations:

- ① Press  once to enter the menu page.
- ② Use  /  to switch between setting pages.
- ③ Press  once to enter the setting page, where the highlighted area appears. Adjust the highlighted numerical value by scrolling the adjustment wheel left or right using .
- ④ After setting is completed, short press  to save and exit.

Menu Items	Adjustable Range	Menu Items	Adjustable Range
Overvoltage Protection	0V~30.00V	Language Setting	Chinese/English
Overcurrent Protection	0A~5.1A	Style Switching	Regular/Industrial
Overpower Protection	0W~150.00W	Brightness Adjustment	Adjustable (Higher level means brighter backlight)
Overheat Protection	0°C~99°C	Volume Control	Adjustable (Higher level means louder volume)
Undervoltage Protection	0V~30V	Measurement Switch	On/Off
About	/		



Menu Interfaces

5.INSTRUCTIONS FOR PC SOFTWARE USE

5.1 Basic Function Interface

- ①Waveform Display Area:Records, stores, and deletes waveform data, including voltage and current. When viewing a single waveform, you can zoom in and out using the mouse scroll wheel.
- ②Output Voltage and Current Settings: Change the output voltage and current values of the current preset group (changes are temporary and not saved; the device will revert to initial values after reboot).
- ③Data Group Data: Click on a data group to set preset voltage and current values. Up to 6 different preset values can be saved.
- ④Measurement Switch: Measures output capacity and energy. Measurements are reset upon closing and reopening.
- ⑤Brightness Settings: 14 levels (Higher level means brighter backlight).
- ⑥USB Interface: Product Model/Firmware Version/Communication Rate/Communication Port/Device Address/Online Status
- ⑦Basic Information: Input Voltage/Output Voltage/Output Current/Output Power/Temperature Display/Status Display

Control Device On/Off Status On This Interface



Basic Function Interface

5.2 Advanced Function Interface

- ①**Sequential Output:** Within the specified range of sequence numbers and loop counts, sequentially output voltage and current parameters.

Click "Start" to execute sequential output (other interfaces are locked and inactive). Numbers that have been executed show "OK", while numbers that haven't been executed show "Waiting". Click "Pause" to maintain the current sequence output. Click "Continue" to execute subsequent steps according to the set delay.

Manual Mode: Click "Single Step" for testing, testing proceeds according to the number of clicks. Stopping will maintain the current sequence output.

- ②**Current Sweep:** With fixed voltage, the current is scanned and output within the set range according to the step current and delay, commonly used in constant current mode. Voltage setting needs to be greater than 0V; Starting current, ending current range: 0.000A5.000A (if the starting and ending values are the same, it's invalid); Step current range: 0.001A5.000A; Delay time: 1s~86400s. Click "Start" to execute current scanning output (other interfaces are locked and inactive). Click "Stop" to stop the scan and close the output.

- ③**Voltage Sweep:** With fixed current, the voltage is scanned and output within the set range according to the step voltage and delay, commonly used in constant voltage mode. Current setting needs to be greater than 0A; Starting voltage, ending voltage range: 00.00V30.00V (if the starting and ending values are the same, it's invalid); Step voltage range: 00.01V30.00V; Delay time: 1s~86400s. Click "Start" to execute voltage scanning output (other interfaces are locked and inactive). Click "Stop" to stop the scan and close the output.



Advanced Function Interface

5.3 PC Software Connection

- ① Turn on the device and connect it to the computer using a Micro data transfer cable.
 - ② After connection, select the communication port. You can view it through Device Manager in Computer Management.
 - ③ Click "Connect". Once "Connected" is displayed in the left text box, the connection is successful.
- ※ **NOTE:** During the PC software connection process, the device buttons will be locked and cannot be operated.

6.FIRMWARE UPGRADE

Method One:

- ① Obtain the latest firmware from the official website and download it to the desktop after decompression.
- ② Press and hold  , then connect the device to the computer using a Micro data transfer cable to enter firmware upgrade mode. At this point, the computer will recognize the device as a USB flash drive.
- ③ Copy the firmware to the USB flash drive. After successful copying, the device will automatically start the firmware upgrade process.
- ④ Observe the upgrade progress percentage. After the upgrade is completed, the device will restart. If the upgrade fails, please contact customer service immediately.

Method Two (Connecting to PC Software):

- ① On the device's About page, ensure that the device address matches the device address displayed in the PC software.
- ② Connect the device to the computer using a Micro data transfer cable. Confirm the port and click "Connect" on the PC software.
- ③ Once the connection is successful, click on "Firmware Upgrade" above, then click "BOOT/Restart."
- ④ The computer will recognize the device as a USB flash drive. Copy the firmware to the USB flash drive. After successful copying, the device will automatically start the firmware upgrade process.
- ⑤ Observe the upgrade progress percentage. After the upgrade is completed, the device will restart. If the upgrade fails, please contact customer service immediately.



Firmware Upgrade Interface

7. PRECAUTIONS

- The product operates in step-down mode, so ensure that the input voltage is higher than the output voltage.
- The power supply range for this product is DC 5V~30V. Output will be disabled if the input voltage is lower than 5.0V due to undervoltage lockout, and the device may be damaged if the input voltage exceeds 30V. It is recommended to use a power supply ranging from 5V to 30V.
- When supplying power to inductive and capacitive loads, it is recommended to connect the load before turning on the device output.
- When outputting high power, some heat generation is normal. It is recommended to use the device in a well-ventilated environment.
- Please remember not to connect power to both power ports of the device simultaneously, as simultaneous power supply may potentially damage the input source with lower voltage.
- When charging batteries, it is strongly recommended to add a reverse current protection module (such as a Schottky diode) to the output end to protect the device from damage.

8.PRODUCTION INFORMATION

Any FNIRSI's users with any questions who comes to contact us will have our promise to get a satisfactory solution +an extra 6 months warranty to thanks for your support! By the way, we have created an interesting community, welcome to contact FNIRSI staff to join our community.

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Received by Email 2025 May 29 - video not yet made on tiktok
follow-up with FNIRSI to post a video on youtube.

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meets commerce



Exciting news – your sample order is now with you! This is your moment to create and captivate your audience. Keep in mind, you have 14 days from today to post your content and start earning commissions on every sale you generate.



 FNIRSI DSO152 Mini Handheld Digital Oscilloscope
2.5MSa/s 200KHz Analog Bandwidth 800VPP Support
Firmware Upgrade PWM Output

Order Number: 576991538312679714

Order Date: 2025-05-27 09:30:18(GMT+0)

Commission Rate: 10%

14 days left to post content with sample product link

FNIRSI™



便携式数控直流电源

PORTABLE CNC DC POWER SUPPLY

FNIRSI™

DPS-150

便携式数控直流电源 | Portable CNC DC Power Supply

生产商 / Manufacturer

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GJB851-1990

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FNIRSI

DPS-150

便携式数控直流电源 | Portable CNC DC Power Supply

产品参数

品牌型号	DPS-150	输出电压	0~30V
输入电压	DC5.0V~32V	输出电流	0~5A
输入电流	100mA~5A	输出功率	0~150W
负载调整率	0.49%	满载效率	96.30%
尺寸	106×76×28mm	重量	≈178g
工作环境	0°C~40°C, 0%~75%RH		
保护机制	●过压保护 ●过流保护 ●欠压保护 ●过功率保护 ●过温保护 ●输入防反接 ●输出防反灌		

注意事项

- 产品工作在降压模式, 需要保证输入电压高于输出电压。
- 本产品供电范围DC5V~30V, 低于5.0V会欠压锁定禁止输出, 超过30V则可能会损坏设备。建议使用5V~30V电源供电。
- 当设备给感性负载和容性负载供电时, 建议先接好负载, 再开启设备输出。
- 当进行大功率输出时, 会有一定程度的发热, 这是正常现象, 建议在通风良好的环境使用
- 设备的两个供电口切记不可同时接电, 同时供电有概率会使电压低的输入源损坏
- 用于电池充电时, 强烈建议在输出端加上防倒灌模块(比如肖特基二极管)以保护设备不被损坏

PRODUCT PARAMETERS

Model	DPS-150	Output Voltage	0~30V
Input Voltage	DC5.0V~32V	Output Current	0~5A
Input Current	100mA~5A	Output Power	0~150W
Load Regulation	0.49%	Full Load Efficiency	96.30%
Dimension	106×76×28mm	Weight	≈178g
Operating Environment	0°C~40°C, 0%~75%RH		
Protection Mechanisms	●Overvoltage Protection ●Overcurrent Protection ●Undervoltage Protection ●Overpower Protection ●Overtemperature Protection ●Reverse Input Protection ●Reverse Output Protection		

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